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AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 211 — Past Questions & Materials

by Renance

Course: CSC 211

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 3

Levels: 200

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Paper 1 of 3 — CSC 211

Level 200 • Second Semester • PastQuestion

GOOD DAY!

MY NAME IS IBHATE BLESSED AIZONOFE,
A 400 LEVEL COMPUTER SCIENCE
STUDENT.

KNOWLEDGE IS A LIFE-LONG PURSUIT
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Data Structures and Algorithm

(Multiple Choice Questions)

1. The memory address of the first element of an array is called:
 - a. Floor address
 - b. Foundation address
 - c. First address
 - d. Base address
2. The two key measures to find efficiency of an algorithm are:
 - a. Time and space
 - b. Capacity and Complexity
 - c. Data and space
 - d. Processor and memory
3. The space factor when determining the efficiency of algorithm is measured by:
 - a. Counting the minimum memory needed by the algorithm
 - b. Counting the maximum memory needed by the algorithm
 - c. Counting the average memory needed by the algorithm
 - d. Counting the maximum disk space needed by the algorithm e.
4. Which of the following cases does not exist in the complexity theory?
 - a. Best case
 - b. Worst case
 - c. Average case
 - d. Null case
5. Which of the following data structures is not linear data structure?
 - a. Linked lists
 - b. Array

- c. Both (a) and (b)
 - d. None of the above
6. The operation of processing each element in the list is known as:
- a. Sorting
 - b. Traversal
 - c. Merging
 - d. Inserting

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7. Linked list are best suited:
- a. For relatively permanent collections of data
 - b. For the size of the structure and the data in the structure are constantly changing
 - c. Both (a) and (b)
 - d. None of the above
8. You have to sort a list L consisting of a sorted list followed by a few 'random' elements. Which of the following sorting methods would be especially suitable for such a task?
- a. Bubble sort
 - b. Selection sort
 - c. Quick Sort
 - d. Insertion Sort
9. Which of the following is false?
- a. Tree is a non-linear data structure
 - b. A tree contains a cycle
 - c. A tree with n nodes contains (n-1) edges
 - d. A tree is a connected graph
10. Which two of the following are equivalent for an undirected graph G?
- i. G is a tree
 - ii. There is atleast one path between any two distinct vertices of G
 - iii. G contains no cycles and has (n-1) edges
 - iv. G has n edges
- a. (i) and (ii)
 - b. (i) and (iii)
 - c. (i) and (iv)
 - d. (ii) and (iii)
11. Queue is alist
- a. LIFO
 - b. LILO
 - c. FILO
 - d. FIFO
12. In a circular linked list

a. Components are linked in random manner

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b. There is no beginning and no end

c. Components are arranged hierarchically

d. Forward and backward traversal within the list is permitted

13. If $(\text{rear} = \text{maxsize} - 1) \text{ rear} = 0$; else $\text{rear} = \text{rear} + 1$; is required in Circular queue

Stack

c. Linear queue

d. Link list

14. The maximum degree of any vertex in a simple graph with n vertices is:

a. $n+1$

b. $n-1$

c. $2n-1$

d. n

15. The data structure required for breadth first traversal on a graph is:

a. Queue

b. Stack

c. Array

d. Tree

16. The quick sort algorithm exploitdesign technique

a. Greedy

b. Dynamic programming

c. Backtracking

d. Divide and conquer

17. The data structure required to evaluate a postfix expression

a. Queue

b. Stack

c. Array

d. Linked list

18. Preorder is also known as

a. Depth first order

b. Breadth first order

c. Topological order

d. Linear order

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19. The data structure required to check whether expression contains balanced parenthesis is

a. Queue

b. Array

c. Tree

d. stack

20. Which of the following data structure is most efficient in terms of both space and time to reverse a string of characters?

a. Linked list

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b. Stack

c. Array

d. tree

21. If a sequence of operations- push(1), push(2), pop, push(1),push(2),pop, pop , pop, push(2),pop are performed on a stack , the sequence of popped out values are

a. 2,2,1,1,2

b. 2,2,1,2,2

c. 2,1,2,2,1

d. 2,1,2,2,2

22. Which of the following is useful in implementing quick sort?

a. Recursion

b. Breadth first search

c. Depth first search

d. Both (a) and (c)

23. How many elements are present in the root of the tree?

a. 1

b. 3

c. 2

d. 4

24. A linear collection of data, where element is given by means of a pointer, is called

a. Linked list

b. Primitive list

c. Code list

d. None of these

25. Which of the following is true for walk in a graph?

a. Edges are not repeated

b. Edges may be repeated

c. All nodes may be covered

d. None of the above

26. In what tree, for every node the height of its left subtree and right subtree differ atleast by one?

a. Binary search tree

b. AVL tree

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- c. Threaded binary tree
 - d. Complete tree
27. A graph is a tree, if and only if it
- a. Is completely connected
 - b. Is minimally connected
 - c. Contains a circuit
 - d. Is planar
28. In any undirected graph, the sum of degrees of all the nodes
- a. Must be even
 - b. Are twice the number of edges
 - c. Must be odd
 - d. Need not even
29. A graph in which all nodes are of equal degree is called
- a. Multi graph
 - b. Non-regular graph
 - c. Regular graph
 - d. Complete graph
30. A simple graph in which there exists an edge between every pair of vertices is called
- a. Complete graph
 - b. Euler graph
 - c. Planar graph
 - d. Regular graph
31. Two isomorphic graphs must have
- a. Equal no. of vertices
 - b. Same no. of edges
 - c. Same no. of vertices
 - d. All of the above
32. Stack is a:
- a. FIFO
 - b. LIFO
 - c. LILO
 - d. LIFO
33. In, the elements must be in sorted order
- a. Linear search
 - b. Quick sort
 - c. binary search
 - d. selection sort
34. The structure which is best suited for hierarchical relationship is:
- a. circular queue
 - b. tree

c. graph

d. stack

35. Which of the following require extra memory for storage:

a. Linked list

b. Array

c. Both (a) & (b)

d. None of the above

36. In which of the following link field of last node point to the first node

a. singly linked list

b. doubly linked list

c. singly circular linked list

d. both (a) and (c)

37. In which traversal root node is visited at the last

a. Post-order traversal

b. Pre-order traversal

c. In-order traversal

d. None

38. is a condition when there is no space to insert an element

a. underflow

b. houseful

c. memory-full

d. overflow

39. Which data structure is used to implement a recursive function call?

a. linked list

b. binary tree

c. stack

d. queue

40. is the process of arranging the elements of a particular data structure in some logical order.

a. Merging

b. Insertion

c. traversing

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d. Sorting

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Multiple Choice Questions for Data Structures

1. Minimum number of fields in each node of a doubly linked list is ____

- (A) 2
- (B) 3
- (C) 4
- (D) None of the above

Ans: B

3

2. A graph in which all vertices have equal degree is known as ____

- (A) Complete graph
- (B) Regular graph
- (C) Multi graph
- (D) Simple graph

Ans: A

Complete graph

3. A vertex of in-degree zero in a directed graph is called a/an

- (A) Root vertex
- (B) Isolated vertex
- (C) Sink
- (D) Articulation point

Ans: C

Sink

4. A graph is a tree if and only if graph is

- (A) Directed graph
- (B) Contains no cycles

- (C) Planar
- (D) Completely connected

Ans: B

Contains no cycles

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5. The elements of a linked list are stored

- (A) In a structure
- (B) In an array
- (C) Anywhere the computer has space for them
- (D) In contiguous memory locations

Ans: C

Anywhere the computer has space for them

6. A parentheses checker program would be best implemented using

- (A) List
- (B) Queue
- (C) Stack
- (D) Any of the above

Ans: C

Stack

7. To perform level-order traversal on a binary tree, which of the following data structure will be required?

- (A) Hash table
- (B) Queue
- (C) Binary search tree
- (D) Stack

Ans: B

Queue

8. Which of the following data structure is required to convert arithmetic expression in infix to its equivalent postfix notation?

- (A) Queue
- (B) Linked list
- (C) Binary search tree
- (D) None of above

Ans: D

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None of above

9. A binary tree in which all its levels except the last, have maximum numbers of nodes, and all the nodes in the last level have only one child it will be its left child. Name the tree.

- (A) Threaded tree
- (B) Complete binary tree

(C) M-way search tree

(D) Full binary tree

Ans: B

Complete binary tree

10. Which of following data structure is more appropriate for implementing quick sort iteratively?

(A) Deque

(B) Queue

(C) Stack

(D) Priority queue

Ans: C

Stack

11. The number of edges in a complete graph of n vertices is

(A) $n(n+1)/2$

(B) $n(n-1)/2$

(C) $n^2/2$

(D) n Ans: B $n(n-1)/2$

12. If two trees have same structure and but different node content, then they are called ____

(A) Synonyms trees

(B) Joint trees

(C) Equivalent trees

(D) Similar trees

Ans: D

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Similar trees

13. If two trees have same structure and node content, then they are called ____ (A) Synonyms trees

(B) Joint trees

(C) Equivalent trees

(D) Similar trees

Ans: C

Equivalent trees

14. Finding the location of a given item in a collection of items is called

A. Discovering

B. Finding

C. Searching

D. Mining Ans. C searching

15. The time complexity of quicksort is

A. $O(n)$

B. $O(\log n)$

C. $O(n^2)$

D. $O(n \log n)$

Ans. D

$O(n \log n)$

16. Quick sort is also known as

A. merge sort

B. tree sort

C. shell sort

D. partition and exchange sort

Ans. D

partition and exchange sort

17. sorting is good to use when alphabetizing a large list of names.

A. Merge

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B. Heap

C. Radix

D. Bubble Ans. C

Radix

18. The total number of comparisons in a bubble sort is

A. $O(n \log n)$

B. $O(2n)$

C. $O(n^2)$

D. $O(n)$ Ans. A

$O(n \log n)$

19. form of access is used to add and remove nodes from a queue.

A. LIFO, Last In First Out

B. FIFO, First In First Out

C. Both a and b

D. None of these

Ans. B

FIFO, First In First Out

20. New nodes are added to the of the queue.

A. Front

B. Back

C. Middle

D. Both A and B

Ans. B

Back

21. The term push and pop is related to

A. Array

B. Lists

C. Stacks

D. Trees

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Ans. C

Stacks

22. Which of the following is an application of stack?

A. finding factorial

B. tower of Hanoi

C. infix to postfix

D. all of the above Ans. D all of the above

23. The operation of processing each element in the list is known as

A. sorting

B. merging

C. inserting

D. traversal Ans. D traversal

24. The situation when in a linked list START=NULL is

A. Underflow

B. Overflow

C. Houseful

D. Saturated Ans. A

Underflow

25. Which of the following are two-way lists?

A. Grounded header list

B. Circular header list

C. Linked list with header and trailer nodes

D. List traversed in two directions

Ans. D

List traversed in two directions

26. Which is the pointer associated with the availability list?

A. FIRST

B. AVAIL

C. TOP

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D. REAR

Ans. B

AVAIL

27. Which of the following data structure can't store the non-homogeneous data elements? A) Arrays

B) Records

C) Pointers

D) Stacks Ans. A

Arrays

28. Which of the following is non-linear data structure?

- A) Stacks
- B) List
- C) Strings D) Trees Ans. D

Trees

29. To represent hierarchical relationship between elements, which data structure is suitable?

- A) Dequeue
- B) Priority
- C) Tree
- D) Graph

Ans. C

Tree

30. Identify the data structure which allows deletions at both ends of the list but insertion at only one end. A) Input restricted dequeue

- B) Output restricted dequeue
- C) Priority queues
- D) Stack

Ans. A

Input restricted dequeue

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Paper 3 of 3 — CSC 211 TEST QUESTIONS

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AMBROSE ALLI UNIVERSITY, EKPOMA

FACULTY OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

2021/2022 SECOND SEMESTER CONTINUOUS ASSESSMENT

CSC 211: DATA STRUCTURES AND ALGORITHM

Instruction: Attempt ALL Questions Time: 20 minutes

1(a) In your understanding, discuss the reasons for data structure

(b) List and discuss five types of data structure.

2(a) Discuss five characteristics of an Algorithm

(b) Write an algorithm to find the largest among three numbers.

3(a) Discuss an array using relevant examples.

(b) List five applications of arrays

4 With diagrams, discuss the various types of queues.

5. With relevant examples, discuss the Stack terminology.